

Modular Power Electronics Approach for High-Power Dynamic Wireless Charging System

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Abstract—Dynamic wireless power transfer (DWPT) can provide energy to EVs in motion and extend the driving range. Upscaling the charging power to 200 kW (High Power DWPT) reduces the percentage of electrified roadway required, and the solution becomes cost-effective. To smooth the power at the battery and grid, a secondary regulation stage must be added. The DWPT system, therefore, relies on power electronics to interface with the coils and regulate the power flow. Designing this high-power system using wide-bandgap devices makes ensuring high efficiency, small size, and reliable operation very challenging, and significant engineering effort is required to build such complicated systems for large-scale installation and deployment. This article describes a modular design approach for power electronics to achieve the 200-kW wireless power transfer (WPT). As described, the silicon-carbide (SiC) power electronics building block is designed, simulated, and characterized. The approach is validated in the DWPT system to build the inverter, the rectifier, and the dc/dc converter, which demonstrated high performance and reliable operation with 188-kW power.

Index Terms—Building block, dynamic wireless power transfer (DWPT), electric vehicle, modular, silicon-carbide (SiC).

I. INTRODUCTION

WIRELESSLY charging EVs is a convenient alternative to conductive charging and is of increased appeal due to the push toward autonomous vehicles generally, including EVs. Dynamic wireless power transfer (DWPT) can provide energy to EVs while they are in motion, thereby extending the driving range and eliminating or significantly mitigating range anxiety in a cost-effective way [1], [2], [3]. Additionally, dynamic charging can reduce the required onboard battery capacity, thereby reducing the weight, volume, and cost of EVs. The DWPT power should be sufficiently high to reduce the total required coverage of the electrified roadway but not high enough to increase the cost of vehicle-side electronics.

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Charging power levels in the range of 200 kW are considered necessary to minimize the percentage of roadway to be electrified while significantly mitigating range anxiety [4]. Not significantly exceeding 200 kW ensures that the vehicle cost does not increase due to the very high charge rates and the associated battery and power electronic costs.

When in motion over the partially electrified roadway, the vehicle's receiver coil receives intermittent electric power, which must be controlled for proper battery charging. Primary-side control can adjust power transmission via inverter duty cycle, phase shift, or simply inverter input voltage [5]. Primary-side control can ensure a constant coil current for safe operation but cannot fulfill the battery charging profile [6]. Instead, a secondary-side regulation scheme with fast feedback should be implemented [7]. This can be achieved by adding a dc/dc conversion stage as a postregulator to enforce the battery charging profile. The overall system architecture is depicted in Fig. 1, which shows both the wireless power transfer (WPT) coils and the power electronics, including the inverter, rectifier, and dc/dc converter, all in multiphase configurations to handle the 200-kW power.

Although inductive power transfer for dynamic wireless charging has been extensively reviewed in [8], high-power (i.e., >200 kW) WPT is still rare. Both Bombardier and Momentum Dynamics have press releases regarding 200-kW DWPT for electric buses [9], [10], but very limited technical details have been provided. Development of 1-MW inductive power transfer has been reported for trains, but the article did not focus on power electronics development [11].

Power electronics design for such a high-power DWPT system is challenging, especially for wide-bandgap devices [e.g., silicon carbide (SiC)]. To achieve lightweight and compact designs, wide-bandgap devices should be used to switch at a high frequency to enable the use of small passive components, including inductors and capacitors, and smaller heat sinks for cooling. Designing systems with robust switching and low power loss becomes challenging in this case because of the complicated power architecture used for power processing, protection, and control. According to the literature, WPT research using a modular power electronics approach has been limited in output power (i.e., <50 kW) [12], [13]. The Power Electronics Building Block approach [14] packages the power functionality and a control interface in a modular, standardized fashion to reduce engineering effort and ease

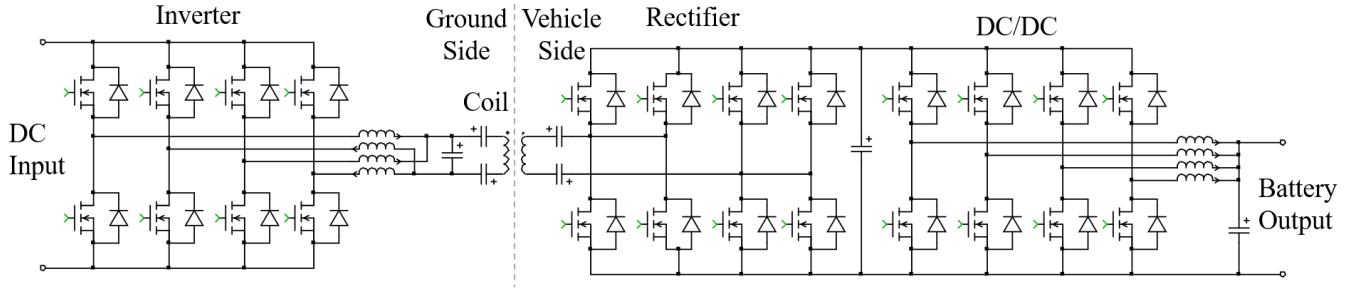


Fig. 1. Overall power architecture of the high-power dynamic wireless charging system.

upscaling challenges. The concept was recently implemented with SiC power devices for multilevel converters [15].

This article presents a modular design approach with a focus on the current scaling capability for a 200-kW DWPT system while ensuring reliable and efficient switching of fast SiC devices. Section II addresses the special power electronics requirements for a DWPT system. Section III describes the design details of the building blocks. Section IV shows the coil design details, and Section V shows the experimental validation results. Section VI concludes the article.

II. POWER ELECTRONICS SYSTEM REQUIREMENT

The overall DWPT system assumes an 800-V dc input, either from a rectified output from the ac grid or directly from a local dc grid, and charges a light-duty battery at 300–400 V.

Both inductor–capacitor–capacitor (*LCC*) and series tuning are widely used in WPT systems and can be used on both transmitter and receiver sides. *LCC* tuning on the primary side leads to a constant current source behavior on the primary-side coil at resonance, and the coil current is independent of load and coupling coefficient variations [16]. The ground-side resonant tuning network is the *LCC* type. The constant coupler current behavior of *LCC* tuning lends itself well to DWPT applications because the coils can be energized without the load (i.e., vehicle) present. This allows the vehicle to extract the most amount of energy from the system. However, *LCC* tuning at the secondary side can quickly overcharge the capacitor at the output of the rectifier if the output reduces its power demands (e.g., if the battery management system is disconnected), which can cause an over-voltage failure of the rectifier and the dc/dc stages. In contrast, a series-tuning method exhibits voltage source-like behavior at resonance [17]. The rectifier output voltage is limited to a value well below the voltage rating of the 1200 V SiC semiconductors and the dc capacitors. Therefore, series tuning at the secondary side was selected. Table I summarizes the system specifications and key parameters for the WPT part.

A. Inverter Power Stage Requirement

Based on the operating conditions, three possible SiC phase-leg modules were identified: CAB450M12XM3 (1200 V, 450 A), CAB400M12XM3 (1200 V, 400 A), and CAS325M12HM2 (1200 V, 325 A). For those power modules, power loss can be modeled in PLECS by using I–V curves and switching energy tables at different temperatures. Electric–thermal coupled simulation can accurately model

TABLE I
WIRELESS POWER SYSTEM KEY PARAMETERS

Parameters	Unit	Values	Comments
Input Voltage	V	800	
Output Battery Voltage	V	300–400	
Output Power	kW	200	
Maximum Charging Current	A	500	
WPT Frequency	kHz	85	
Transmitter Coil Inductance	μH	7.1	
Receiver Coil Inductance	μH	22.8	
Coupler Mutual Inductance	μH	1.7	Under aligned condition
Coupler Distance	inch	9	From transmitter surface to receiver surface
Transmitter Side Tuning Inductance	μH	2.2	All four inductors are equal
Transmitter Side Tuning Capacitance (Parallel)	μF	2.38	
Transmitter Side Tuning Capacitance (Series)	μF	0.67	Equivalent overall series capacitance
Receiver Side Tuning Capacitance	μF	0.38	Both capacitors are equal

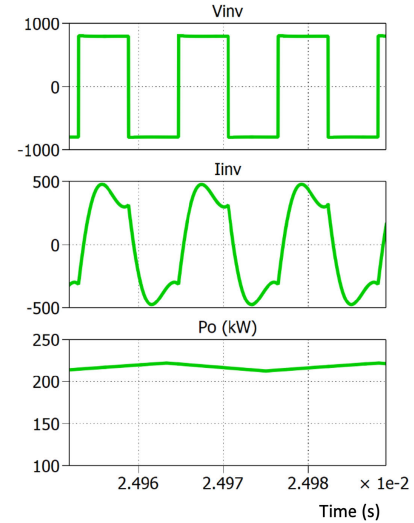


Fig. 2. Simulated waveforms. (Top) V_{inv} —inverter output voltage. (Middle) I_{inv} —inverter phase current. (Bottom) P_o —load power.

switch losses at the thermal steady state with an in situ heat sink. Fig. 2 shows simulated waveforms for the inverter output voltage and phase current at a load power of 218 kW. The phase current has a root mean square (rms) value of 354 and 314 A at the instants of a turn-off.

Based on simulation and analytical studies, a loss analysis was performed for approaches using different modules and paralleled numbers. Notably, Table II shows that having two modules in parallel leads to better efficiencies overall. More importantly, with the single-module approach, each module was close to the thermal power dissipation limit, thereby rendering the single-phase approach infeasible. The junction temperature of the SiC MOSFETs can be calculated by $T_j = T_c + P_{\text{loss}} \times (R_{\text{th,jc}} + R_{\text{th,TIM}} + R_{\text{th,HS}})$, where the three R_{th} symbols correspond to the thermal resistance from the power module, $R_{\text{th,jc}} = 0.1 \text{ } ^\circ\text{C/W}$; the thermal interface material, $R_{\text{th,TIM}} = 0.014 \text{ } ^\circ\text{C/W}$; and the liquid-cooled heat sink,

TABLE II
LOSS ANALYSIS RESULTS OF THE INVERTER STAGE

	CAB450M12XM3		CAB400M12XM3		CAS325M12HM2	
Modules in Parallel	1	2	1	2	1	2
Total Loss (kW)	4.144	3.879	2.571	2.025	2.299	1.821
Inverter Efficiency	98.3%	98.4%	98.9%	99.1%	99.0%	99.2%

TABLE III
LOSS ANALYSIS RESULTS OF THE RECTIFIER STAGE

Part Number	GB2X100MPS12-227	APTDF400U120G	CAS325M12HM2
Technology	SiC schottky	Si p-n junction	SiC MOSFET(Diodes)
Junction Temperature	125°C	100°C	125°C
Quantity	8	4	4
Total Footprint (in ²)	12.9	31.6	44.3
Losses (W)	1.792	1.512	1.350

$R_{th,HS} = 0.02$ °C/W. According to the loss modeling in Table II, the single-phase method will have 1.15-kW power loss per module, whereas the two-phase solution's power loss is only 450 W, which leads to a junction temperature, T_j , of 214 °C and 120 °C, respectively, assuming a 60 °C coolant temperature, T_c . Therefore, this approach was selected for the two-phase design.

B. Rectifier Stage Design

First, different diode bridge options are compared using different technologies, including SiC Schottky, Si p-n-junction, and the integrated diodes of the SiC power module, and the results are shown in Table III. The CAS325M12HM2 power module provides the lowest power loss when using four modules in parallel. The power module's antiparallel diodes are used for the rectifier mode. A CAS325M12HM2-based solution was selected because it offers the ability to use the MOSFETs to switch actively for protection or rectification in the future. In the synchronous rectification mode, the rectifier's power loss reduces from 1350 to 434 W, which is a 68% reduction.

C. DC/DC Stage Design

To achieve an optimized design, parameters including power module selection, switching frequency, and number of interleaved phases should be carefully evaluated. Similar to the modeling process of the inverter design, three off-the-shelf SiC power modules—CAB450M12XM3, CAS325M12HM2, and CAS120M12BM2—with different on-resistance and switching energy are compared. For each power module, its power loss is modeled in PLECS by using I - V curves and switching energy tables at different temperatures. Electric-thermal coupled simulation can accurately simulate switch losses at a thermal steady state with an in situ heat sink. The comparison and design details of this dc/dc converter were reported in [18], and the findings suggest that the CAS325M12HM2 power module achieves the best balance between switching and conduction loss for this application.

Assuming a 60 °C coolant temperature, Fig. 3 plots the resultant junction temperature at different switching frequencies and phase numbers when using CAS325M12HM2. The junction temperature increases almost linearly with switching frequency. Owing to the design, the ripple current stays constant regardless of the switching frequency. This means that as frequency increases, the selected inductance will reduce

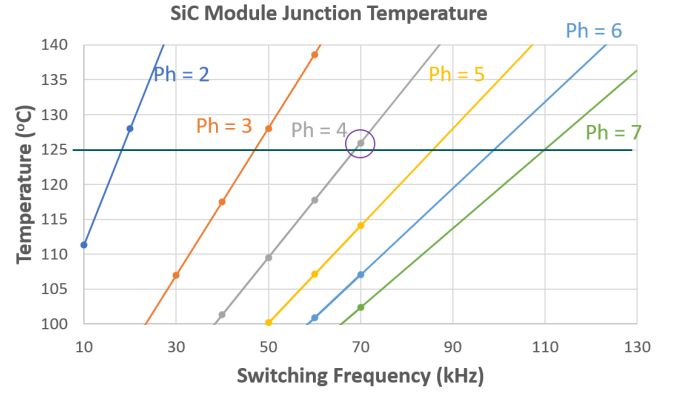


Fig. 3. Simulated SiC module junction temperature.

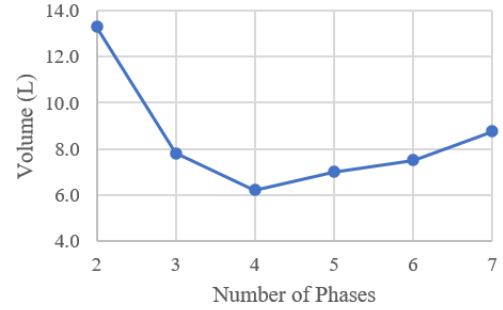


Fig. 4. Overall dc/dc converter volume with a different number of phases, including power modules, gate drivers, heat sinks, and inductors.

proportionally, and so the projected inductor size will reduce. A constant current ripple dominates the switching current for the power semiconductor at turn-on and turn-off, and these will be the same value (as the switching energy E_{on} and E_{off}). As a result, the switching loss, $P_{sw} = (E_{on} + E_{off}) \times f_{sw}$, will be proportional to the switching frequency, f_{sw} . For design cases in which switching loss dominates the conduction loss, the overall power loss will also show a close-to-linear increase with switching frequency. Because the thermal design is also fixed, it is unsurprising that the junction temperature also increases in a similar manner.

By adopting more phases, power losses are shared by multiple power modules; therefore, the converter's thermal management has a higher power loss budget before it hits the targeted device junction temperature (e.g., 125 °C). Therefore, a design with more phases can switch at a higher frequency, which benefits the size reduction of the inductor. For example, a four-phase configuration can operate at 70 kHz, whereas a seven-phase counterpart can operate at 110 kHz, according to Fig. 3. However, at higher phase numbers, the size of the power modules and heat sink increases, which reduces the benefit of inductor shrinking. Fig. 4 shows that a four-phase configuration provides the smallest overall dc/dc converter volume. Therefore, for the dc/dc converter, four-phase and 70-kHz switching frequency is selected.

III. POWER ELECTRONICS BUILDING BLOCK DESIGN

Based on the evaluation from Section II, a building block of a half-bridge or a full bridge with the SiC module CAS325M12HM2 should be designed. The building block should achieve high switching performance with low loss, low

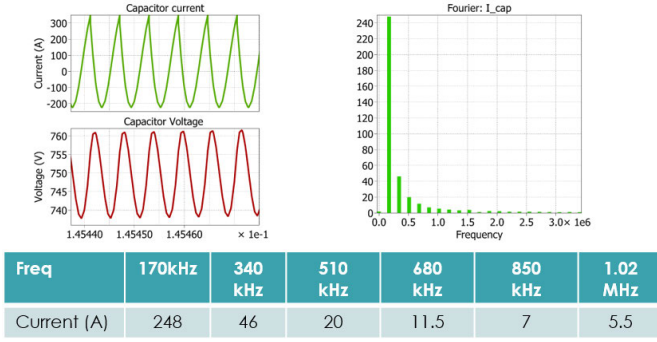


Fig. 5. Simulated capacitor current, voltage, and capacitor current harmonics.

overshoot, low electromagnetic interference (EMI), and good thermal capability while also being scalable for higher power and more phase quantity.

A. Building Block Capacitor Selection and Placement

The SiC modules' high-frequency operation and faster switching transition times lead to smaller WPT couplers and high-efficiency operation, respectively. High-performance decoupling capacitors should be selected and placed as near to the module as possible with minimal impedance between the capacitor and the module to supply the energy required for the high-frequency switching (i.e., ≈ 85 kHz) with short switching transitions (i.e., < 50 ns). A detailed simulation was conducted to estimate the magnitude and frequency component of the current ripple that the decoupling capacitor could supply for efficient and safe 85-kHz operation. As shown in Fig. 5, the decoupling capacitor must be able to supply a large ripple current (≈ 300 A peak) for 250-kW operation. It also shows a considerable amount of current at the higher frequency, including 5.5 A at 1.02 MHz. This indicates that a high-frequency capacitor with a self-resonant frequency above 1 MHz is required. Therefore, a B58033I9505M001 5- μ F, 1300-V lead lanthanum zirconate titanate (PLZT) capacitor [19] with a self-resonant frequency greater than 1 MHz and high ripple current capability was selected as the decoupling capacitor for the power electronics building block.

The PLZT decoupling capacitor exhibits an asymmetrical pin-out pattern in which the outer two rows of four pins represent one electrical terminal, and the center row with two pins serves as the other terminal (see Fig. 6). With a two-layer bus bar configuration, the most intuitive placement is to use the outer four pins as one terminal and the center two pins are the other. Notably, the capacitor can still work identically under reverse-rated voltage [19]. This creates another capacitor placement scheme in which the outer and inner pins are used alternatively for Vdc+ and Vdc-. Because the adjacent capacitors use terminals alternatively, Vdc+ and Vdc- connections are evenly distributed, and this scheme is deemed an *interleaved* scheme in Fig. 6, and the other one is designated as *non-interleaved*. A finite-element analysis (FEA) simulation is conducted in COMSOL Multiphysics 6.0 [20] to compare the performance of the two schemes. Each copper layer is excited by a 1-A current flowing from the source to sink points, as indicated in the figures. The color maps show the voltage potential at the bus bar copper surface when the sink points

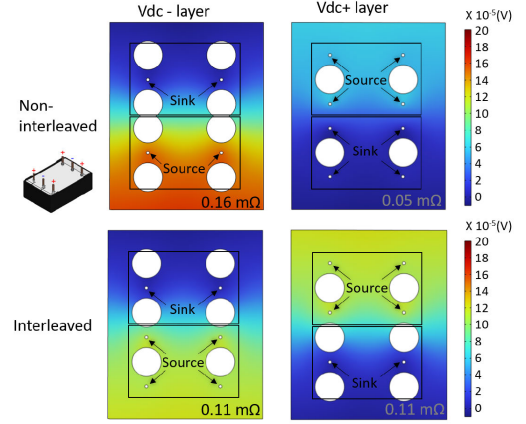


Fig. 6. FEA simulation of the voltage potential of two capacitor placement.

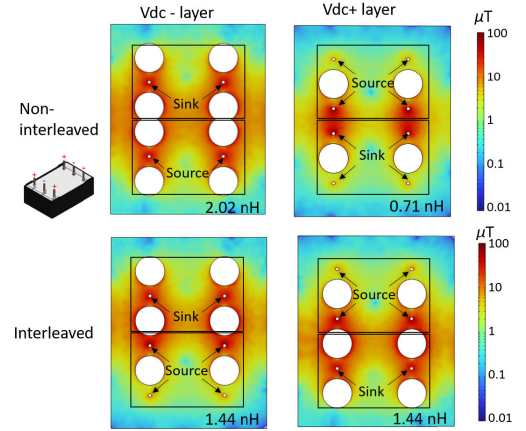


Fig. 7. FEA simulation of the magnetic fields generated by the two-capacitor placement.

were set as zero potential. The noninterleaved scheme has a significantly asymmetrical voltage potential distribution for the two copper layers. The extracted resistance is 0.16 m Ω for the Vdc- layer and 0.05 m Ω for the Vdc+ layer. In contrast, the interleaved structure shows a balanced voltage potential for the two copper bus bar layers, with identical extracted resistance of 0.11 m Ω . Although the total loop resistance is similar (≈ 0.22 m Ω) for both solutions, the asymmetrical distribution will lead to different power loss for the two bus bar layers, and this difference can create mismatched heat expansion and long-term reliability issues. Simulation results in Fig. 7 also show that the interleaved structure has a more symmetrical magnetic field distribution. The extracted partial inductance for the interleaved placement is also balanced (both 1.44 nH) for the two bus bar layers.

Notably, the proposed idea for capacitor placement is not limited to the specific capacitor selected in this work. For any capacitor type that has fail-short characteristics, two or more capacitors must be connected in series in the system so that one capacitor failure would not trigger a short-circuit of the main dc voltage rail. This is particularly important in automotive applications for which high reliability is required. Moreover, these capacitors have internal series-parallel construction to reduce failures caused by ceramic cracks (i.e., from thermal and mechanical stress) compared to multilayer ceramic capacitors.

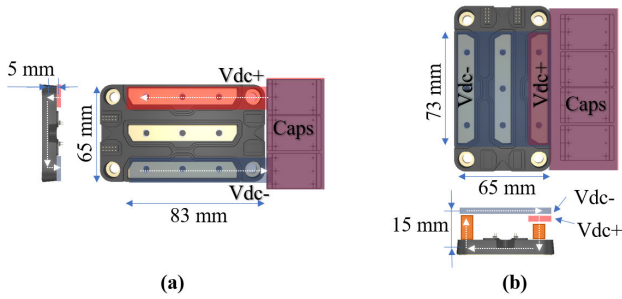


Fig. 8. Comparison of the two bus bar designs. (a) Horizontal switch loop design with top and side views. (b) Vertical switching loop design with top and front views.

B. Printed-Circuit-Board (PCB)-Based Bus Bar Design

The most critical parameter for switching performance is the power switching-loop inductance, which is determined by all interconnect conductors and traces between the power module and the decoupling capacitors. This switching-loop inductance dominates voltage overshoot, switching loss, and EMI and should be minimized. According to the pin out of the selected SiC half-bridge power module, two designs are proposed and compared, as shown in Fig. 8. The first design [see Fig. 8(a)] adopts two copper strips (each is 11 mm wide) for the positive and negative dc rails (Vdc+ in red and Vdc- in blue). The switch node's bus bar will be a similar shape and reside in the middle terminal but is not explicitly shown because it will not impact the switching-loop inductance. With sufficient local decoupling capacitance, the switching loop can be determined as illustrated by the white-dashed lines. It starts from the Vdc+ terminals of the capacitors and the Vdc+ bus bar, travels inside the power module, and then returns to the Vdc- terminal of the capacitors through the Vdc- bus bar. Because most of this loop is in parallel with the module's top surface, this first bus bar option is regarded as a *horizontal-loop* design.

The other design [see Fig. 8(b)] adopts a *vertical-loop* concept. This design's dc bus bars (Vdc+ and Vdc-) are much wider (73 mm) than the horizontal design (11 mm). Because the switch node's bus bar must again connect at the middle terminal of the power module, two copper spacers must be added to separate the two dc bus bars from it in the vertical dimension. With the decoupling capacitors placed at the side, the switching current loop (white-dashed lines in Fig. 8) starts with the Vdc+ bus bar and spacer, continues with the power module and Vdc- spacer, and then returns to the capacitor along the Vdc- bus bar. Because this power loop is completed in a plane that is perpendicular to the module's top surface, it is regarded as a *vertical-loop* design.

Both designs were simulated for the loop inductance in COMSOL Multiphysics 6.0 [20]. The simulated models for both cases are shown in Fig. 9. Design details at the capacitor side (e.g., through holes and copper patterns) can be ignored for comparison purposes because these details contribute similarly to the overall loop inductance in both designs. For the horizontal-loop design, the excitation terminals are selected as the edge surfaces of the Vdc+ and Vdc- bus bar strips. With the two bus bars, the switching loop is completed with the internal electric conductor pattern of the power module, which is assumed to be in a flat U-shape with the dimension of

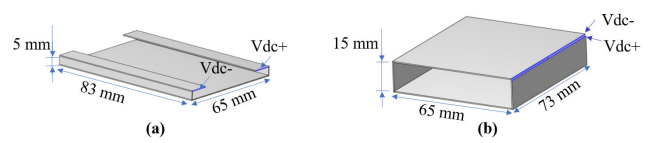


Fig. 9. Simulation models in COMSOL for the two designs. (a) Horizontal loop. (b) Vertical loop. The blue areas with arrows and Vdc+ and Vdc- labels indicate the excitation ports.

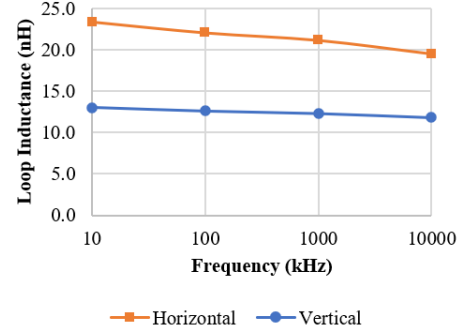


Fig. 10. Extracted switching loop inductance for both horizontal and vertical designs.

$65 \times 83 \times 5$ mm. This approximates the internal layout pattern of the power module according to the literature [21] and should, therefore, represent the switching current conduction path for the loop-inductance extraction. The power module's internal interconnect inductances (e.g., from bond wires) are not captured with this approximation but will not impact the collateral comparison between the two design cases because of equal contribution. The vertical-loop design assumes a 10-mm height for the spacer and a 5-mm height for the flat U-shape structure of the power module, which adds up to 15 mm of height to the overall structure. The excitation terminals are applied at the narrow openings of the enclosed structure to match the design shown in Fig. 8(b).

The simulated loop inductance in both cases is compared in Fig. 10. The vertical-loop design has a lower loop inductance (11.6 nH) than the horizontal-loop design (19.6 nH), which exhibits a 41% reduction. Therefore, the vertical-loop design was selected as the building design. Simulation indicated that the loop inductance of the vertical design linearly reduces with the height of the structure. For example, the loop inductance drops to 4.6 nH if the height is 5 mm.

C. Building Block Construction

Traditionally, power modules are connected to the dc side by a copper bus bar structure with screwed-in bypass capacitors (usually film capacitors). However, the selected PLZT capacitors with the through-hole package are incompatible with the conventional screw-in assembly process. Therefore, in this high-power building block design, a PCB-based approach is used, and the decoupling capacitors can be assembled following a standard reflow or wave soldering process. Mature PCB fabrication also enables a more precise, delicate, and optimized copper pattern manufactured at a low cost. The PCB-based bus bar has two 14 oz. copper layers designated as Vdc- and Vdc+ and shown in Fig. 11. The two copper layers are separated by a 0.4-mm thick FR4 insulation layer to sustain up to 1200 V. Eight PLZT capacitors are placed in two rows

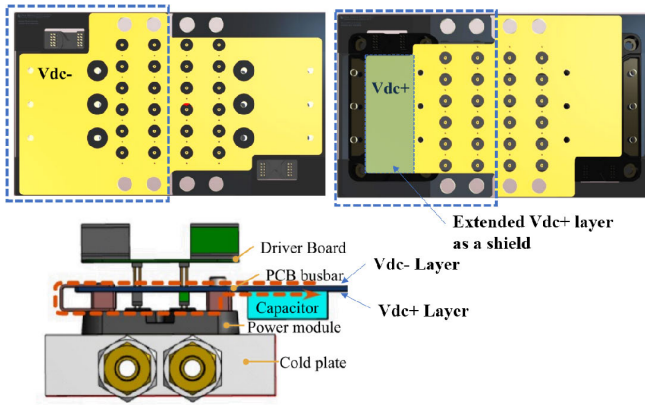


Fig. 11. Two-layer PCB layouts (top two figures) and the building block side view (bottom figure) demonstrate the vertical loop. The side view only shows the left half of the building block because the right half mirrors the left half.

in an interleaved manner to ensure symmetrical resistance and inductance for Vdc+ and Vdc- layers. With the capacitors assembled in the middle, two half-bridge SiC power modules can be accommodated at both sides to form a full-bridge configuration. This design puts the dc+ and dc- terminals at the board edge, and the power rating can be scaled up by cascading the building blocks via those terminals.

To achieve the vertical power loop discussed in Section III-B, a U-shape bus bar is used to interface with the screw-in connection of the power module's negative terminal. A copper spacer bridges the positive terminal of the power module with the Vdc+ copper layer of the bus bar PCB. Both the U-shape connector and the spacer are built as low profile as possible to reduce the loop area and loop inductance. The profile limit comes from the vertical clearance with the switch node's bus bar, which connects in the middle. A 10-mm spacer was selected, which equals the minimum distance between the power terminals of the module, so that the overall minimum electrical clearance stays the same as that of the original power module. The bottom subplot of Fig. 11 shows the completed construction of the building block with PCB bus bar, capacitors, power module, driver board, and cold plate. Notably, only the left half of the building block is shown because the right half simply mirrors the left half. A clear vertical commutation loop is achieved. The loop inductance of this design is simulated in COMSOL. Similar to the analysis in Section III-B, the power module is assumed to be in a flat U-shape with a dimension of $65 \times 83 \times 5$ mm and attached to the bus bar structures. The sink and source are applied at the capacitor pins. Simulated *B*-field distribution in COMSOL is shown in Fig. 12, and the extracted inductance is 12 nH.

The benefit of the proposed structure is switching noise self-containment, which ensures that the sensitive driving and control circuits are contained within an electromagnetically quiet environment. There are three types of noise coupling: conductive, capacitive, and inductive. All three can interfere with the gate drive and signal processing circuits and further corrupt the signals or falsely trigger the pulswidth modulation (PWM) signals to cause shoot-through and other catastrophic failures. With integrated low equivalent series

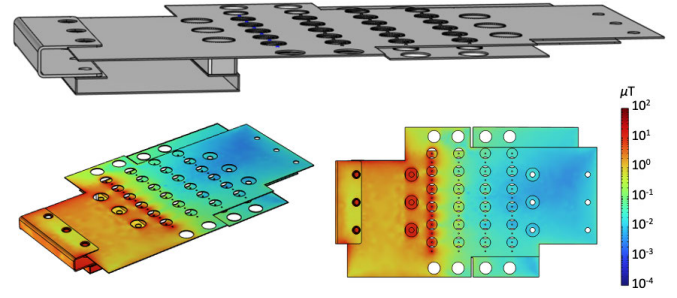


Fig. 12. FEA simulation model and *B*-field distribution in COMSOL.

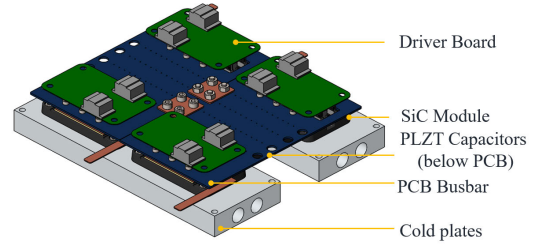


Fig. 13. DC/AC inverter units using two building blocks. The dimension of $306 \times 290 \times 71$ mm indicates a power density of 31.7 kW/L.

inductance (ESL), equivalent series resistance (ESR), and PLZT capacitors, most of the switching current is contained within the building block module and will not conductively propagate to other circuits through a common ground. The main source of capacitive coupling noise is the power module's switch node terminal, which endures a continuous voltage potential change at a high switching frequency and a high slew rate. In the proposed structure, the switch node terminal is shielded by the Vdc+ and Vdc- copper layers of the PCB bus bar, and both layers are terminated to the quiet potentials of the capacitor terminals. Notably, the Vdc+ copper layer is intentionally extended to cover the power module. The extended portion of the Vdc+ layer will be a particularly effective shield because it carries no switching current. For inductive coupling, the switching current loop is the main source of noise because the magnetic flux generated through the loop can induce a voltage in a sensitive circuit. With a minimized current loop, this inductive coupling is also minimized. This design ensures a noise-free condition for the driver board, and the board can be installed on top of the power circuit and still maintain reliable operation.

D. Modular Design Examples

With the fast switching current self-contained, additional building blocks can be easily cascaded for higher power paralleling or multiple phase configurations. The upscaling is achieved by connecting the dc terminals together with interconnect structures. According to the modeling in Section II, the transmitter-side inverter requires two full bridges, so two of the designed building blocks were used to construct the prototype. Fig. 13 shows the CAD model of this design. The overall dimensions are $306 \times 290 \times 71$ mm, and the volume is 6.29 L, which leads to a power density of 31.7 kW/L. Two copper interconnects are used to bolt the adjacent building

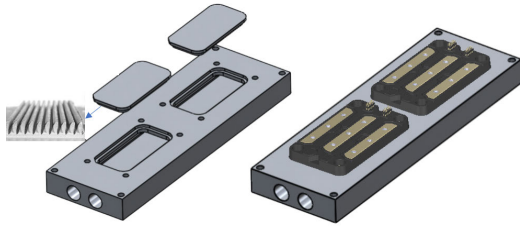


Fig. 14. Customized cold plate design using microcooling channels.

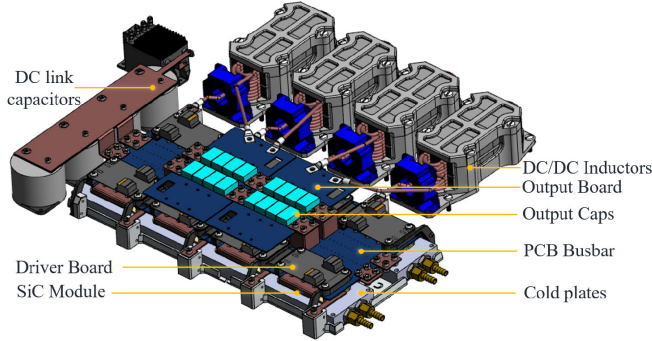


Fig. 15. Rectifier and dc/dc combo unit using four building blocks. The dimension of $541 \times 650 \times 86.5$ mm indicates a power density of 6.6 kW/L.

blocks together. A customized cold plate from Wieland Micro-cool is installed with the two SiC modules on each side. Compared to the standard construction in the literature [22], the installation position was rotated 90° to match the power module orientation, as shown in Fig. 14. The cold plate adopts Micro Deformation Technology's [22] patented inline pin fin (up to 20 pins/in), friction-stir welded construction, and computational fluid dynamics-optimized parallel cooling to achieve close to 0.02°C/W of thermal resistance.

Both the secondary rectifier and the dc/dc will need four half bridges for optimal performance. This two-stage solution can be implemented with four full-bridge building blocks because they share the common dc bus. Fig. 15 shows the CAD model of the rectifier's integrated unit and the four-phase dc/dc converter with the filter components. For easy cable routing, the left four power modules are used for the rectifier, and the right four modules are used for the dc/dc buck converter so that the receiver coils can be fed to the left of the assembly, and buck inductors are placed on the right. Note that two more building-block PCB boards equipped with higher capacitance ($20\ \mu\text{F}$) and lower voltage (500 V) PLZT capacitors (TDK B58033I5206M001) are used for the buck output and placed at the top of the unit. More conventional film capacitors are added at the dc link as low-frequency energy storage for control purposes. The unit's cold plates are similar to that of the inverter, except the length is almost doubled to hold four modules instead of two. The overall dimension of $541 \times 650 \times 86.5$ mm indicates a power density of 6.6 kW/L. Note that this volume includes the intermediate film bus capacitors and the necessary sensors for system regulation control.

With multimodule paralleling required, and considering the fast switching speed of SiC power devices, a modular approach can contain the switching current within each switching cell to minimize power-loop inductance and reduce overshoot and EMI. Power levels can be scaled up by cascading more

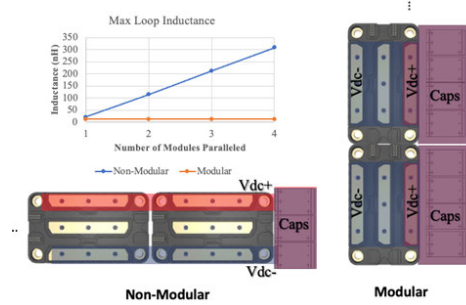


Fig. 16. Loop inductance comparison between modular and nonmodular designs.

modular building blocks, and the loop inductance for each individual power module will stay the same. In contrast, a nonmodular design approach in this case means the power modules will be paralleled directly, so the farthest power module will suffer from high loop inductance because the switching current must travel through a long loop. The maximum loop inductance at the farthest module can rise as high as 308 nH if four power modules are paralleled in a nonmodular approach. This is more than $23\times$ higher than the modular approach (13 nH), as shown in Fig. 16. In the modular design, $di/dt = 85\text{V}/21\text{ nH} = 4\text{ A/ns}$. The peak voltage across the switch when considering voltage overshoot will be $L \times di/dt$. Given the loop inductance of 114 nH with two modules in parallel, the maximum voltage will be 1295 V, which already exceeds the module's voltage rating. This proves the need for a modular design approach in the proposed high-power systems.

IV. WIRELESS COUPLER DESIGN AND ANALYSIS

For stationary wireless charging applications, WPT coils have been designed to optimize the power transfer efficiency over a small range of misalignments. For the dynamic charging scenario, the net energy transferred over the entire overlap window as the vehicle coil moves over the primary coil must be considered. Additionally, for dynamic charging applications, the ground or primary coil must be embedded in a roadway lane and withstand much harsher conditions than in stationary parking lot charging.

A. Circular and Double-D Comparison

Because of the naturally large misalignment in the direction of vehicle travel, shielding the vehicle chassis is an important consideration. Initial design work indicated that the circular coupler topology had better performance characteristics than the double-D topology. However, the vehicle-length shielding requirements affected the performance of the circular coupler. Initial circular coupler power dropped by more than 50% when the vehicle-length shielding was added. However, the double-D coupler power was almost unaffected by the shielding [23]. Design modifications to accommodate shielding significantly decreased the circular coupler advantages. Even after solving the shielding issue, the sensitivity of the circular topology was still a concern with many uncertainties for a practical system. Therefore, the robustness of double-D topology took priority over the reduced mass of the circular topology. There are additional practical reasons to prefer the double-D topology

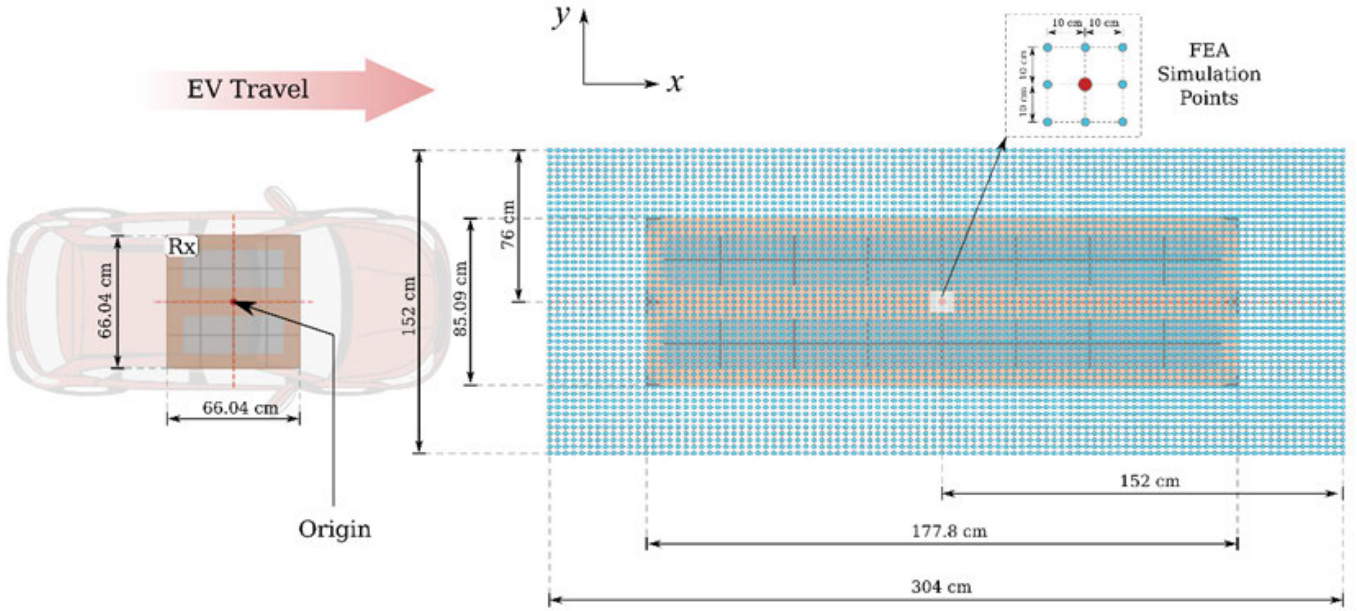


Fig. 17. 200-kW WPT transmitter and receiver coil dimensions with FEA simulation points.

for high-power DWPT systems. To reduce the coupler voltage drop, multiple litz wires can be run in parallel to form the coupler winding while keeping the number of turns low. In the circular topology, the innermost wire would have a lower inductance than the outermost wire. This inductance imbalance would cause a current imbalance that would reduce the system's performance. In the double-D topology, the multiple litz wires are transposed between the two loops of the double-D coil for better inductance and current balance between wires.

B. Optimized Double-D Coil Design

An elongated transmitter coil comparable to the length of a light-duty EV was developed along with a 200-kW vehicle receiver optimized to fit underneath the vehicle. The width of the transmitters had an upper limit of a roadway's lane width. This allows for deploying the DWPT in square precast concrete blocks and provides accommodation for expansion joints if the couplers are directly buried. All ground- and vehicle-side electromagnetic coupler designs were constrained to fit within the smallest area bounded by the tires of the vehicle. The surface of the ground couplers was assumed to be buried 3 in. beneath the surface of the pavement. Assuming a minimum ground clearance of 6 in. for the vehicle couplers, the total coupler surface-to-surface air gap was 9 in. The mutual inductance required to transfer $P_o = 200$ kW at a nominal operating frequency of $f_o = 85$ kHz and limit the coil current to a 750 A peak was determined to be $1.5 \mu\text{H}$ in the following equation:

$$M = \frac{P}{w_o I_{\text{pri}} I_{\text{sec}}} \quad (1)$$

where $w_o = 2\pi f_o$, I_{pri} is the primary coil current, and I_{sec} is the secondary coil current.

C. FEA

The transmitter is 1.78×0.85 m, and the receiver coil is 0.66×0.66 m. Fig. 17 shows the transmitter and receiver

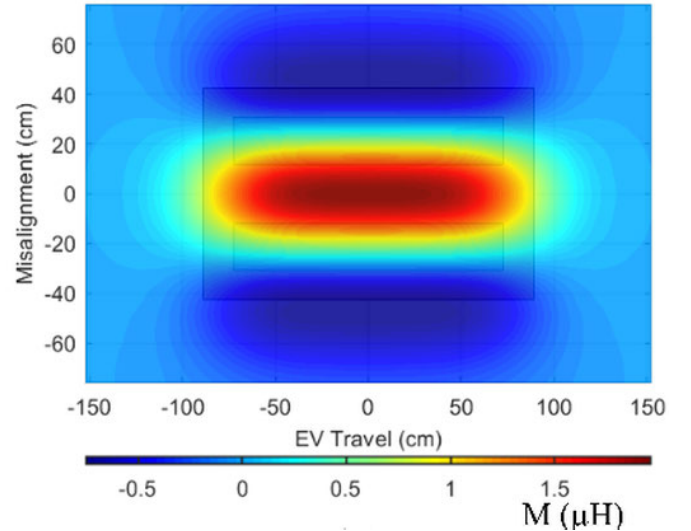


Fig. 18. Variation of mutual inductance as a function of x and y misalignment for the 200-kW DWPT system.

coil dimensions along with the FEA simulation data points. Fig. 18 shows the mutual inductance, M , acquired through FEA simulations over the range of $x = -152$ cm to $x = 152$ cm and $y = -76$ cm to $y = 76$ cm, where the origins of the transmitter and receiver coils are aligned at $x = y = 0$. The air gap between the transmitter and receiver coil is 25.4 cm.

As depicted in Fig. 18, the mutual inductance between the transmitter and receiver coil was the highest at the central limb (variations in x when $y = 0$). The mutual inductance decreased as misalignment along the y -direction increased. The FEA simulation value of the mutual inductance was close to $1.7 \mu\text{H}$, which meets the design target of $1.5 \mu\text{H}$. The output power capability was, therefore, determined and is plotted in Fig. 19, assuming a vehicle dc-link voltage of 800 V and primary-side LCC tuning and secondary-side series tuning.

Thermal constraints also limit the number of ampere-turns on the ground coupler and maximum flux density in the

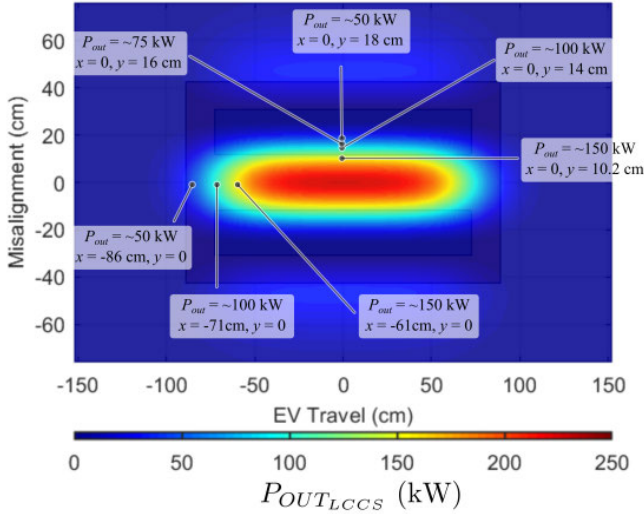


Fig. 19. Variation of output power as a function of x and y misalignment for the 200-kW DWPT system.

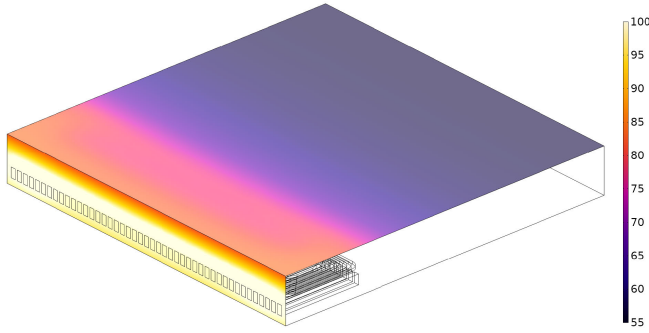


Fig. 20. Thermal analysis of the DWPT system embedded in concrete with an ambient temperature of 55 °C for half of the lane (6 ft width).

ferrites. Fig. 20 shows the simulated steady-state temperature distribution on the pavement surface and through the roadway cross section assuming an ambient temperature of 55 °C. The design was constrained to limit the maximum hot-spot temperature of 100 °C. This simulation assumes 100% uptime for the ground coupler. Fig. 21 shows a map of the critical regions for electromagnetic field emissions just above the surface of the pavement. Half of the 12-ft wide lane is depicted. The transmitter meets the ICNIRP and SAEJ2954 specifications for pacemaker safety of 15 μTrms at the lane boundary. This indicates that high-power DWPT systems can be safely implemented in dedicated highway lanes without causing safety concerns for adjacent vehicles.

V. EXPERIMENTAL VALIDATION

A. Building Block Double Pulse Characterization

The designed building block was first characterized in a double pulse test (DPT) configuration, with the resultant turn-on and turn off waveforms shown in Fig. 22 and Fig. 23, respectively. Per the envisioned operating condition, the upper bound of the MOSFET dc voltage is ≈ 750 V, and the turn-off current is ≈ 250 A. The DPT inductor value and the duty cycle of the bottom MOSFET (device under test) were varied for a turn-off current of 244 A at 750 V dc. The resulting voltage overshoot was 85 V over the 750 V dc link. This is ample margin for the maximum device rating of 1200 V. Additionally,

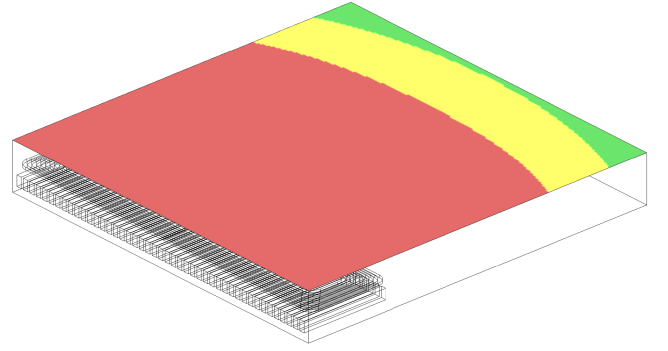


Fig. 21. Map of critical electromagnetic field emission regions on the surface of the roadway: green is below 15 μTrms , yellow is between 15 and 27 μTrms , and red is above 27 μTrms .

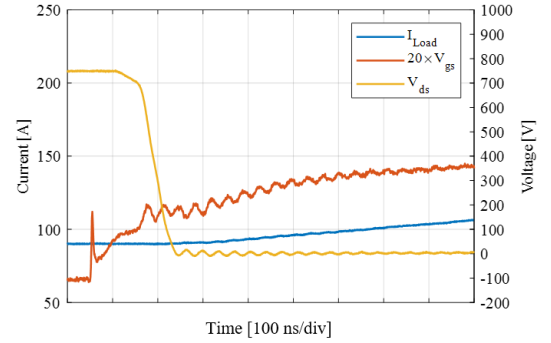


Fig. 22. Turn-on switching waveforms of the building block.

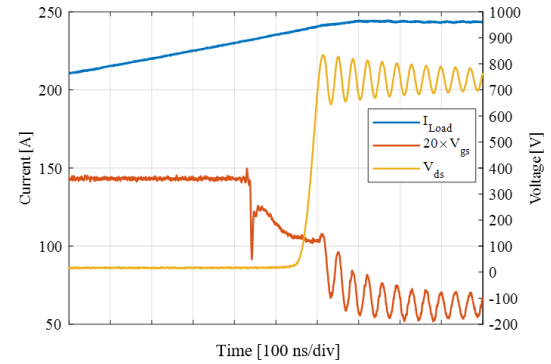


Fig. 23. Turn-off switching waveforms of the building block.

the voltage oscillation period is measured as 36.3 ns, which indicates a loop inductance of 21 nH when considering the SiC power module's output capacitance of 1.57 nF at 750 V dc. The inductance value is higher than extracted due to the unmodeled power module's inductance from the bond wires (5 nH) and the ESL of the capacitance (4 nH), according to the component data sheets.

B. Building Block Validation in a WPT System

A 200-kW WPT system was built using the developed inverter, rectifier, and dc/dc converter. The system specifications and key parameters for the WPT are summarized in Table I.

The transmitter and receiver coils shown in Fig. 17 were assembled using 3C95 ferrite bars, AWG 2-equivalent Litz wire for the transmitter, AWG 4-equivalent Litz wire for the receiver, and additively manufactured wire holders. The

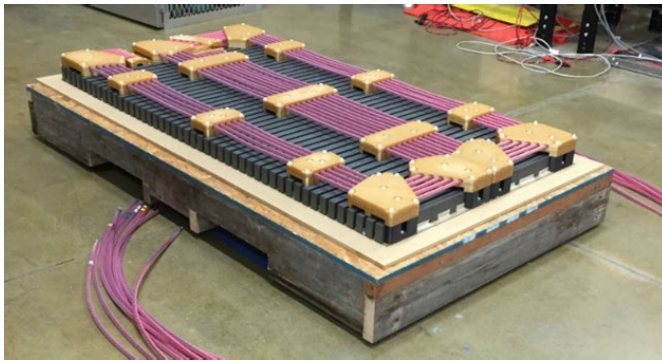


Fig. 24. Fabricated 200-kW transmitter coil.

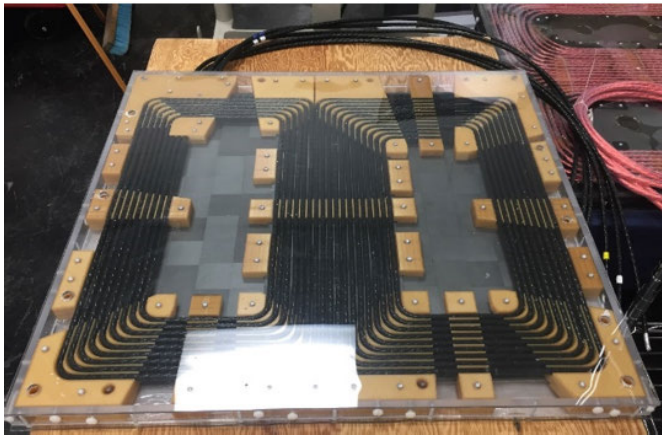


Fig. 25. Fabricated 200-kW receiver coil.

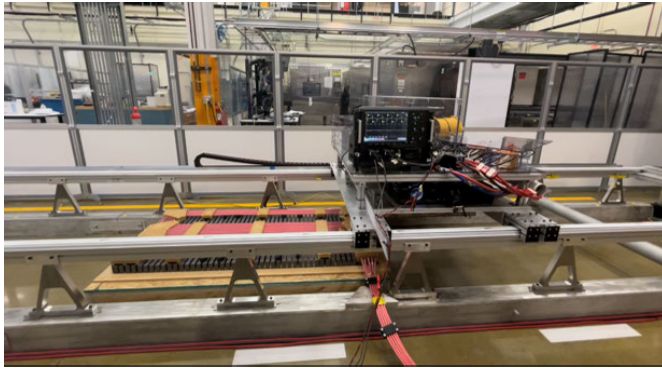


Fig. 26. 200-kW WPT system.

transmitter coil will be assembled, and MasterFlow concrete will be poured over it in a controlled environment to build a precast concrete block embedded with the transmitter coil. The prototypes of the 200-kW ground-side transmitter and receiver coils are shown in Figs. 24 and 25, respectively. The long lead length shown in Fig. 24 was designed to traverse the length of the electrified lane and connect to the ground-side power electronics assembly placed by the roadway.

The three subplots in Fig. 26 show the four-phase inverter, the integrated rectifier and dc/dc converter, and the WPT coil/tuning components. Only one DD transmitter coil is shown to characterize the power transfer capability and efficiency in the aligned condition. A digital platform with a TMS320F28377D digital signal processor (DSP) from Texas Instruments is used to generate the PWM and implement

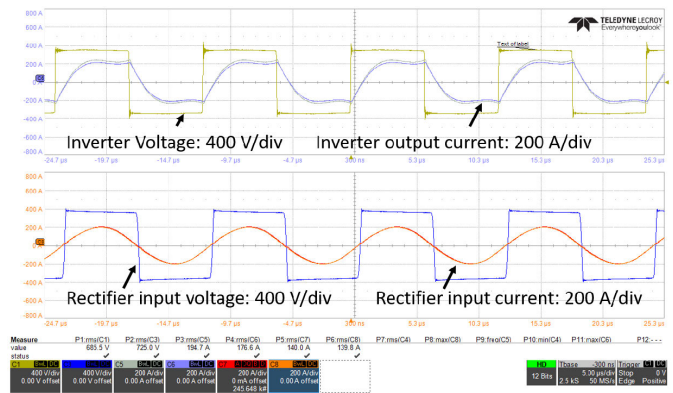
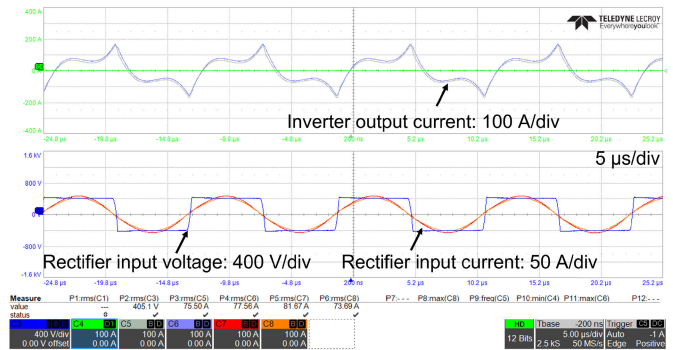


Fig. 27. WPT waveforms tested at 700 V input and 736 V output at 186 kW and aligned condition.

Fig. 28. WPT waveforms tested at 700-V input and misaligned condition at a relative x -axis = 3 (or 0.75 m).

feedback control. Liquid cooling is applied to all the building blocks to dissipate the heat from power loss.

The WPT system was first tested without the dc/dc stage, and the waveforms are shown in Fig. 27. With the designed resonant network slightly inductive, zero-voltage switching is achieved for all the inverter switches. The two full bridges for the dc/ac inverter achieve good current sharing (194.7 versus 176.6 A) at the primary side. The receiver side achieves excellent current sharing (140.0 versus 139.8 A). At 700-V input, the WPT system received 736 V and 186 kW after the rectifier with 93.27% efficiency.

The WPT system was also tested at different alignment conditions. The waveforms are shown at relative x -axis = 3 in Fig. 28 and at relative x -axis = 5 in Fig. 29. Compared with the aligned condition in Fig. 27, the receiver-side current and voltage reduce with misaligned distance. The transferred power and corresponding efficiency are shown in Fig. 30.

Fig. 31 shows the loss breakdown pie charts for the WPT system without the dc/dc converter. Notably, around 11% of the power loss is unaccounted for in the modeling, and this is mainly coming from the additional ac winding losses caused by unmodeled stray magnetic fields, the high-frequency interconnect, and contact resistance losses.

The four-phase dc/dc converter was tested with 800-V input and 400-V output at 188-kW output power. The dc/dc converter was configured to switch at 70 kHz in continuous conduction mode. The waveforms are shown in Fig. 32. The interleaved inductor current is shown with a 90° phase shift between the adjacent phases. Current balancing among all four

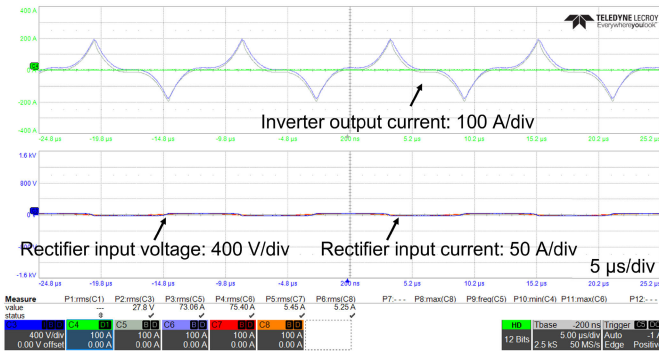


Fig. 29. WPT waveforms tested at 700 V input and misaligned condition at a relative x -axis = 5 (or 1.25 m).

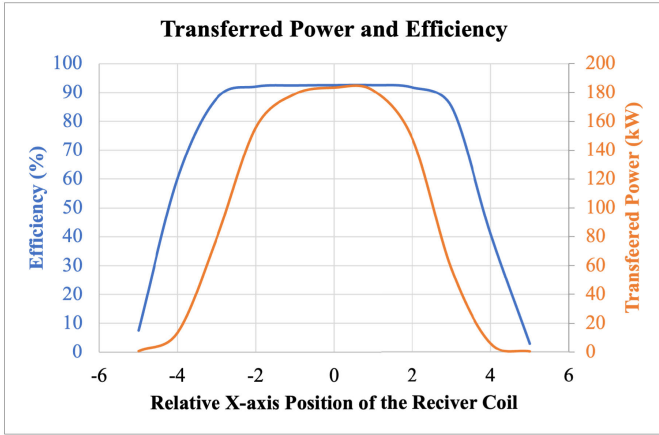


Fig. 30. Measured power and efficiency at different alignment conditions. The horizontal axis is the relative position of alignment, where $x = 0$ corresponds to the aligned condition with a 0.25 m increment in misalignment per position.

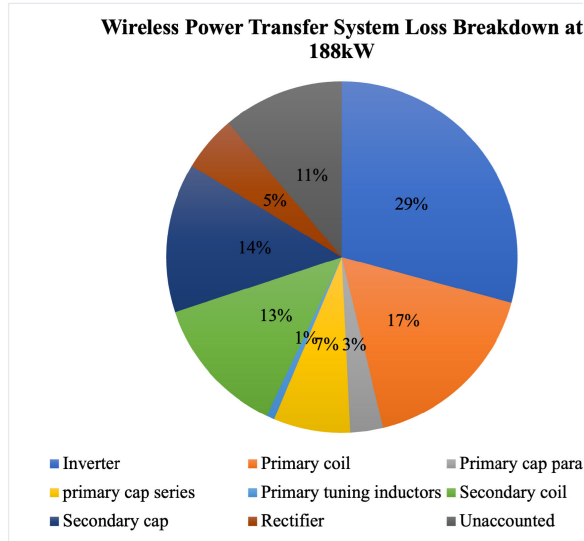


Fig. 31. Loss breakdown of the WPT system at 186 kW.

phases is very well maintained. The measured efficiency at this condition is 98.43%.

Loss breakdown pie charts for the dc/dc converter are shown in Fig. 33. The figure clearly depicts the domination of switching losses in the converter for those high charger devices.

The proposed work is compared with the state-of-the-art WPT systems, as shown in Table IV. Because no WPT

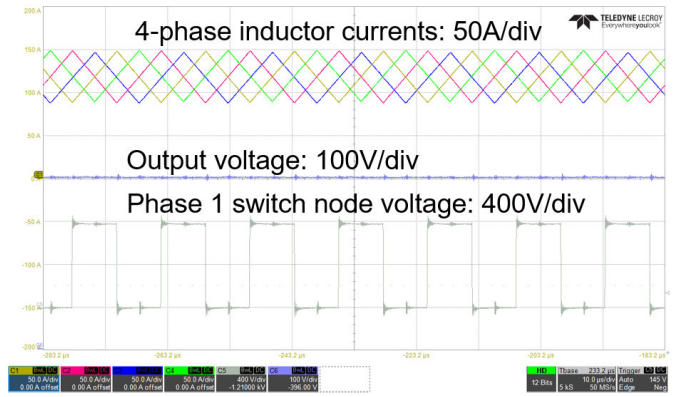


Fig. 32. DC/DC stage waveforms tested at 800-V input and 40-V output at 188 kW.

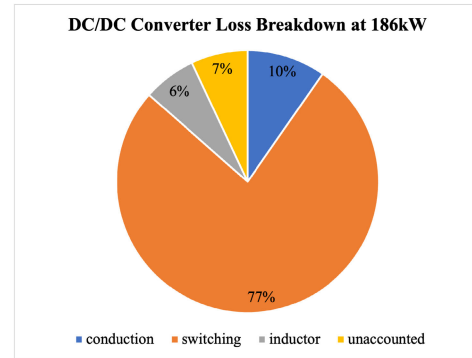


Fig. 33. Loss breakdown of the dc/dc converter at 188 kW.

TABLE IV

COMPARISON OF THIS WORK WITH OTHER EFFORTS IN THE LITERATURE

Parameters	[13]	[12]	[24]	This Work
Dynamic WPT	No	No	Yes	Yes
Modularity	Yes	Yes	No	Yes
Power (kW)	50	35	50	200
Efficiency	96.5%	94%	88%	93.3%
Frequency (kHz)	85	85	25	85
Airgap (mm)	52	200	100–152	230

experiments at 200 kW have been reported, a few lower-power examples are selected for comparison. This work outperforms the state of the art in both transferred power and distance and optimizations to improve efficiency will continue.

VI. CONCLUSION

This article presented a modular approach to designing the power electronics of a 200-kW dynamic wireless charging system for EVs. A full-bridge building block that integrated 1200-V SiC modules, driver boards, low-ESL capacitors, and cold plates was designed with a heavy copper PCB-based bus bar structure. The high-power building block achieved low-loop inductance and low-voltage overshoot. Multiple building blocks are easily cascaded at the dc terminals to form a dual full bridge to drive the ground-side coil and an integrated four-phase rectifier-dc/dc converter. The experimental results show over 93.27% WPT efficiency from the dc input to 186-kW dc output (after the rectifier). The four-phase Buck dc/dc converter was validated with 98.43% efficiency at 188 kW, which leads to an expected overall

system efficiency of approximately 92%. This article shows that the proposed modular approach significantly simplifies the power electronics development effort for a high-power dynamic wireless charging system while achieving consistent and superior performance compared with the state of the art.

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REFERENCES

- [1] A. Foote, O. C. Onar, S. Debnath, M. Chinthavali, B. Ozpineci, and D. E. Smith, "Optimal sizing of a dynamic wireless power transfer system for highway applications," in *Proc. IEEE Transp. Electrification Conf. Expo (ITEC)*, Jun. 2018, pp. 1–6.
- [2] A. N. Azad, A. Echols, V. A. Kulyukin, R. Zane, and Z. Pantic, "Analysis, optimization, and demonstration of a vehicular detection system intended for dynamic wireless charging applications," *IEEE Trans. Transport. Electrification*, vol. 5, no. 1, pp. 147–161, Mar. 2019.
- [3] A. Zaheer, M. Neath, H. Z. Z. Beh, and G. A. Covic, "A dynamic EV charging system for slow moving traffic applications," *IEEE Trans. Transport. Electrification*, vol. 3, no. 2, pp. 354–369, Jun. 2017.
- [4] V. P. Galigekere, "High power and dynamic wireless charging of electric vehicles (EVs)," U.S. DOE Vehicle Technol. Office Annu. Merit Rev. Washington, DC, USA, Tech. Rep. ELT197, 2019. [Online]. Available: <https://www.energy.gov/eere/vehicles/articles/high-power-and-dynamic-wireless-charging-electric-vehicles-evs-0>
- [5] O. C. Onar, S. L. Campbell, L. E. Seiber, C. P. White, and M. Chinthavali, "Vehicular integration of wireless power transfer systems and hardware interoperability case studies," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, Sep. 2016, pp. 1–8.
- [6] R. Tavakoli, A. Jovicic, N. Chandrappa, R. Bohm, and Z. Pantic, "Design of a dual-loop controller for in-motion wireless charging of an electric bus," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, Sep. 2016, pp. 1–8.
- [7] G. Guidi and J. A. Suul, "Transient control of dynamic inductive EV charging and impact on energy efficiency when passing a roadside coil section," in *Proc. IEEE PELS Workshop Emerg. Technol., Wireless Power Transf. (Wow)*, Jun. 2018, pp. 1–7.
- [8] A. Mahesh, B. Chokkalingam, and L. Mihet-Popa, "Inductive wireless power transfer charging for electric vehicles—A review," *IEEE Access*, vol. 9, pp. 137667–137713, 2021.
- [9] *Bombardier's Primove Technology Enters Service on Scandinavia's First Inductively Charged Bus Line—News | Bombardier*. Accessed: Aug. 9, 2022. [Online]. Available: <https://bombardier.com/en/media/news/bombardiers-primove-technology-enters-service-scandinavia-first-inductively-charged-bus>
- [10] (2019). *Momentum Dynamics Fully Automated Inductive Charging for All Types of Electric Buses*. [Online]. Available: <https://www.tesc.psu.edu/assets/docs/momentum-dynamics-wireless-vehicle-charging.pdf>
- [11] J. H. Kim et al., "Development of 1-MW inductive power transfer system for a high-speed train," *IEEE Trans. Ind. Electron.*, vol. 62, no. 10, pp. 6242–6250, Oct. 2015.
- [12] Q. Deng, P. Sun, W. Hu, D. Czarkowski, M. K. Kazmierczuk, and H. Zhou, "Modular parallel multi-inverter system for high-power inductive power transfer," *IEEE Trans. Power Electron.*, vol. 34, no. 10, pp. 9422–9434, Oct. 2019.
- [13] R. Bosshard and J. W. Kolar, "All-SiC 9.5 kW/dm³ on-board power electronics for 50 kW/85 kHz automotive IPT system," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 5, no. 1, pp. 419–431, Mar. 2017.
- [14] T. Ericson, N. Hingorani, and Y. Khersonsky, "PEBB—Power electronics building blocks from concept to reality," in *Proc. Rec. Conf. Papers IEEE Ind. Appl. Soc. 53rd Annu. Petroleum Chem. Ind. Conf.*, Sep. 2006, pp. 1–7.
- [15] J. Wang et al., "Power electronics building block (PEBB) design based on 1.7 kV SiC MOSFET modules," in *Proc. IEEE Electr. Ship Technol. Symp. (ESTS)*, Aug. 2017, pp. 612–619.
- [16] S. Li, W. Li, J. Deng, T. D. Nguyen, and C. C. Mi, "A double-sided LCC compensation network and its tuning method for wireless power transfer," *IEEE Trans. Veh. Technol.*, vol. 64, no. 6, pp. 2261–2273, Jun. 2015.
- [17] V. P. Galigekere, O. Onar, J. Pries, S. Zou, Z. Wang, and M. Chinthavali, "Sensitivity analysis of primary-side LCC and secondary-side series compensated wireless charging system," in *Proc. IEEE Transp. Electrification Conf. Expo (ITEC)*, Jun. 2018, pp. 885–891.
- [18] L. Xue, V. Galigekere, E. Gurpinar, G. Su, and O. Onar, "Modular design of receiver side power electronics for 200 kW high power dynamic wireless charging system," in *Proc. IEEE Transp. Electrification Conf. Expo (ITEC)*, Jun. 2021, pp. 744–748.
- [19] *CeraLink Capacitors for Fast-Switching Semiconductors*, TDK Electron., Bengaluru, India, 2019.
- [20] *Now Available: COMSOL Multiphysics Version 6.0 | COMSOL Blog*. Accessed: Jan. 15, 2022. [Online]. Available: <https://www.comsol.com/blogs/now-available-comsol-multiphysics-version-6-0/>
- [21] *Wolfspeed CAS325M12HM2*, System Plus Consulting, La Chapelle-sur-Erdre, France, Feb. 2019.
- [22] *CP 3002—Wieland MicroCool*. Accessed: Feb. 21, 2022. [Online]. Available: <https://www.microcooling.com/our-products/cold-plate-products/3000-series-cold-plates/cp-3002/>
- [23] M. Mohammad, O. C. Onar, V. P. Galigekere, G. Su, and J. Wilkins, "Magnetic shield design for the double-D coil-based wireless charging system," *IEEE Trans. Power Electron.*, vol. 37, no. 12, pp. 15740–15752, Dec. 2022.
- [24] I. Villar, A. Garcia-Bediaga, U. Iruretagoyena, R. Arregi, and P. Estevez, "Design and experimental validation of a 50 kW IPT for railway traction applications," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, Sep. 2018, pp. 1177–1183.



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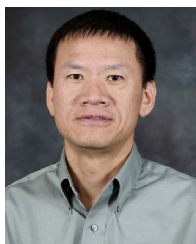
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